LINEAR REGRESSION

Learning Objectives

- Recognise when a linear regression is an appropriate method to use
- Interpreting y = mx + c
- Evaluate linear models

Expected Prior Knowledge

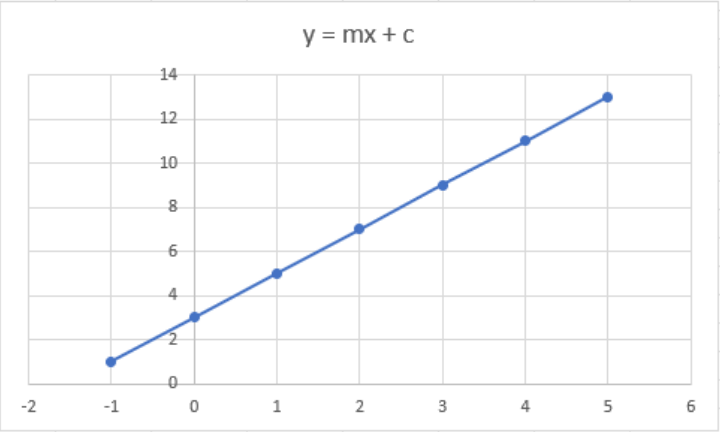
- Statistics for Data Analysis modules: Introduction to Hypothesis Testing

EQUATION OF A LINE AND TERMINOLOGY

All straight lines have an equation that can be written in the form y = mx + c

x	y = mx + c
-1	1
0	3
1	5
2	7
3	9
4	11
5	13

m:	2
c:	3



m is referred to as the **coefficient of x**. On a plot we see its effect as the **gradient** of the line. For Data Analytics: the symbol $\boldsymbol{\beta_1}$ might be used instead. We can interpret it as the impact that changing x has on the predicted y value – if x increases by 1 unit, m is the amount that the y prediction changes by.

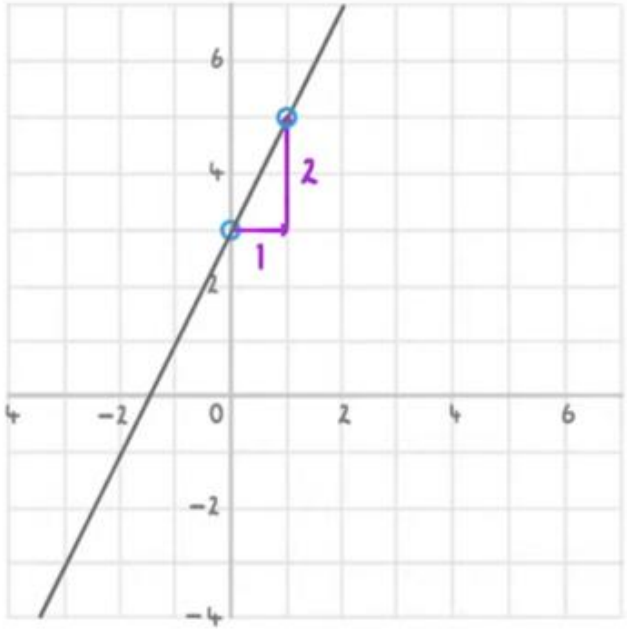
c is a **constant**. On a plot it controls the height of the graph – when x=0 we see that it is where the line **intercepts the y-axis**. For Data Analytics the symbol $\boldsymbol{\beta_0}$ might be used instead.

EQUATION OF A STRAIGHT LINE

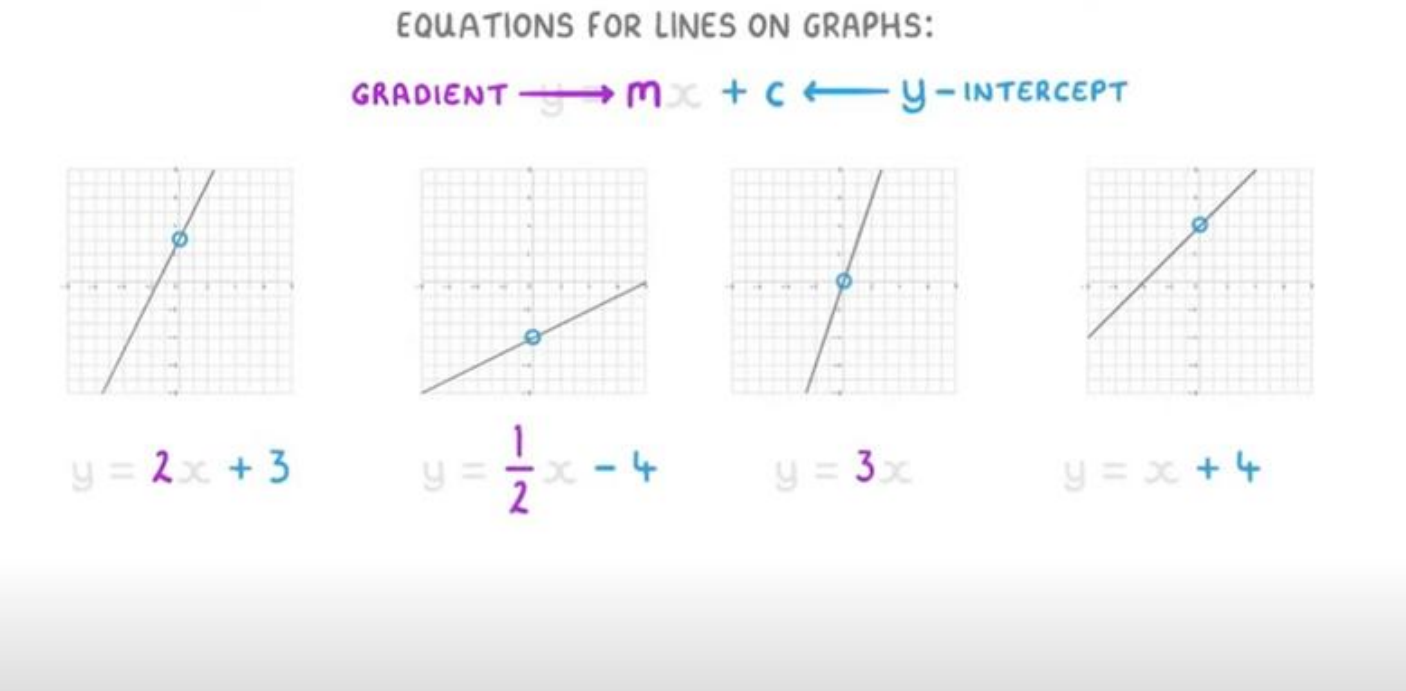
EQUATIONS FOR LINES ON GRAPHS:

$$y = mx + c$$

$$y = 2x + 3$$



EQUATION OF A STRAIGHT LINE



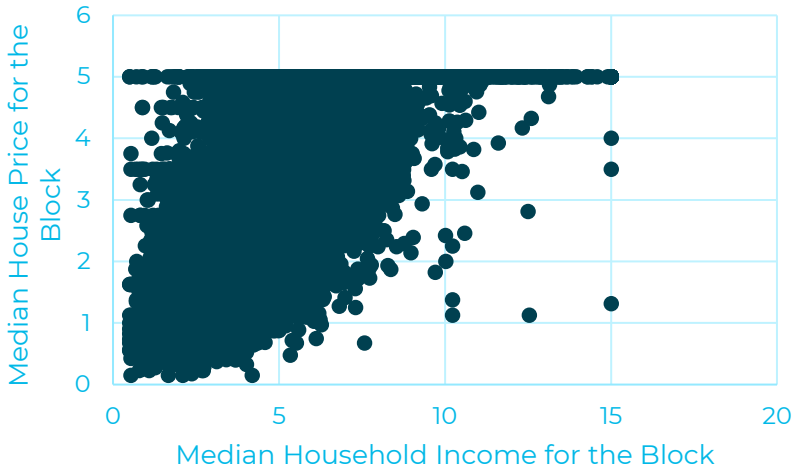
GOAL: PREDICTING Y

Linear Regression is used for **predicting a continuous variable**, y.

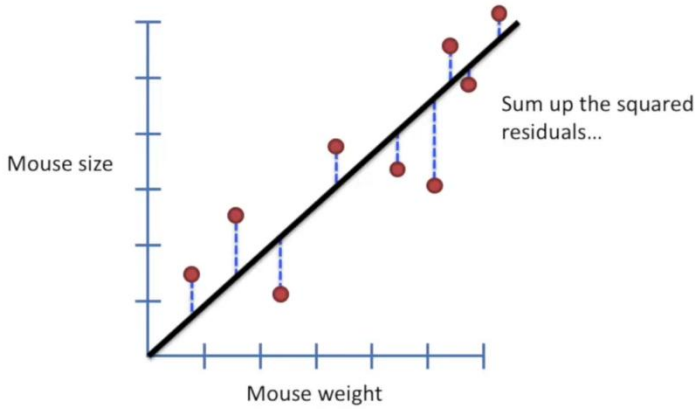
We ask the question: what variable might it be useful to predict?

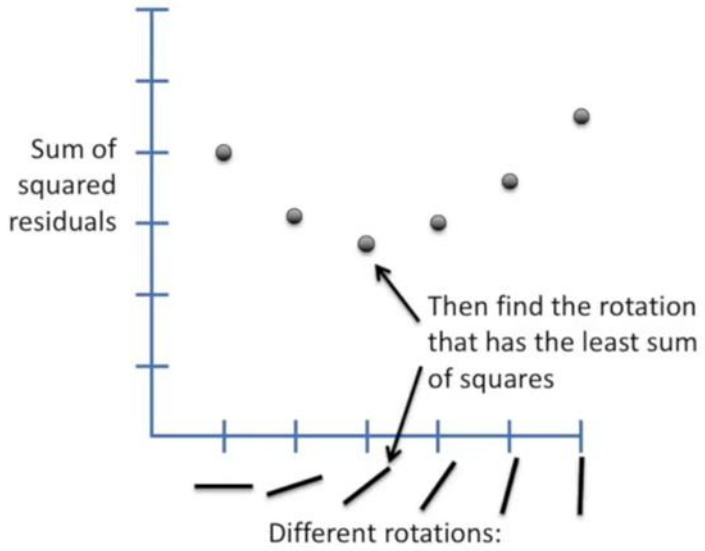
This variable can be placed on the vertical axis of a scatter plot and we can try different variables on the x axis to work towards predicting y accurately.

Example: Predicting the average house price for a ‘block’ in California:



LINEAR REGRESSION





DISCUSSION:

POTENTIAL USE CASES

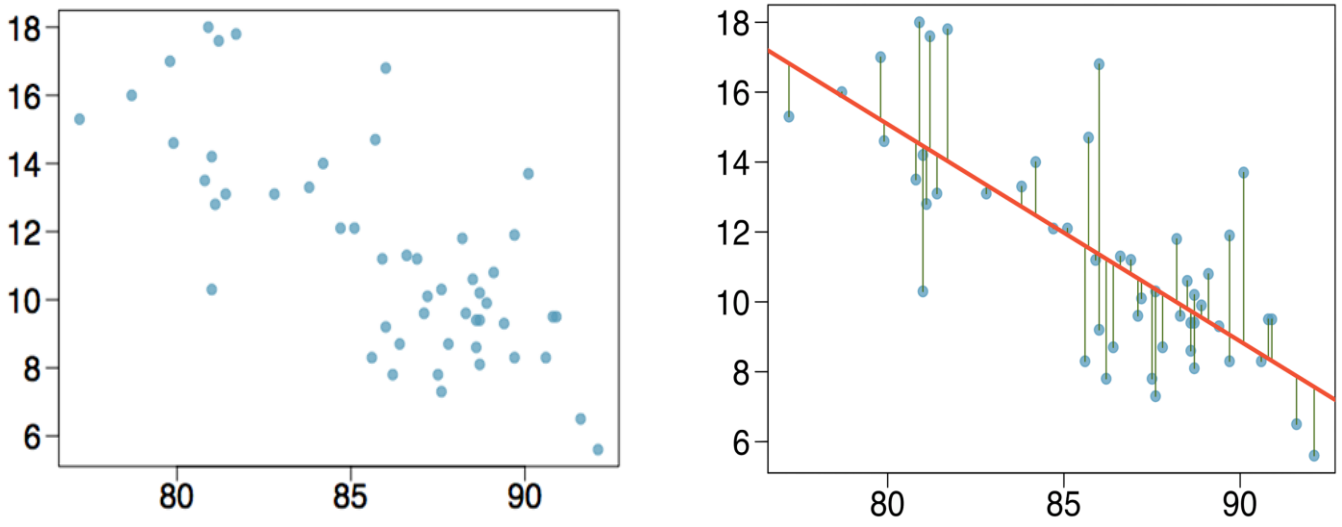
What might it be useful to predict?

What other variables do you suspect might be connected?

Note: If you find yourself thinking of examples where you want to predict ‘yes’ or ‘no’ rather than a variety of continuous numbers – you will be interested in **classification models** rather than linear regression. Make a note of your ideas for later use.

RESIDUALS

Real data is rarely located in perfect straight lines!



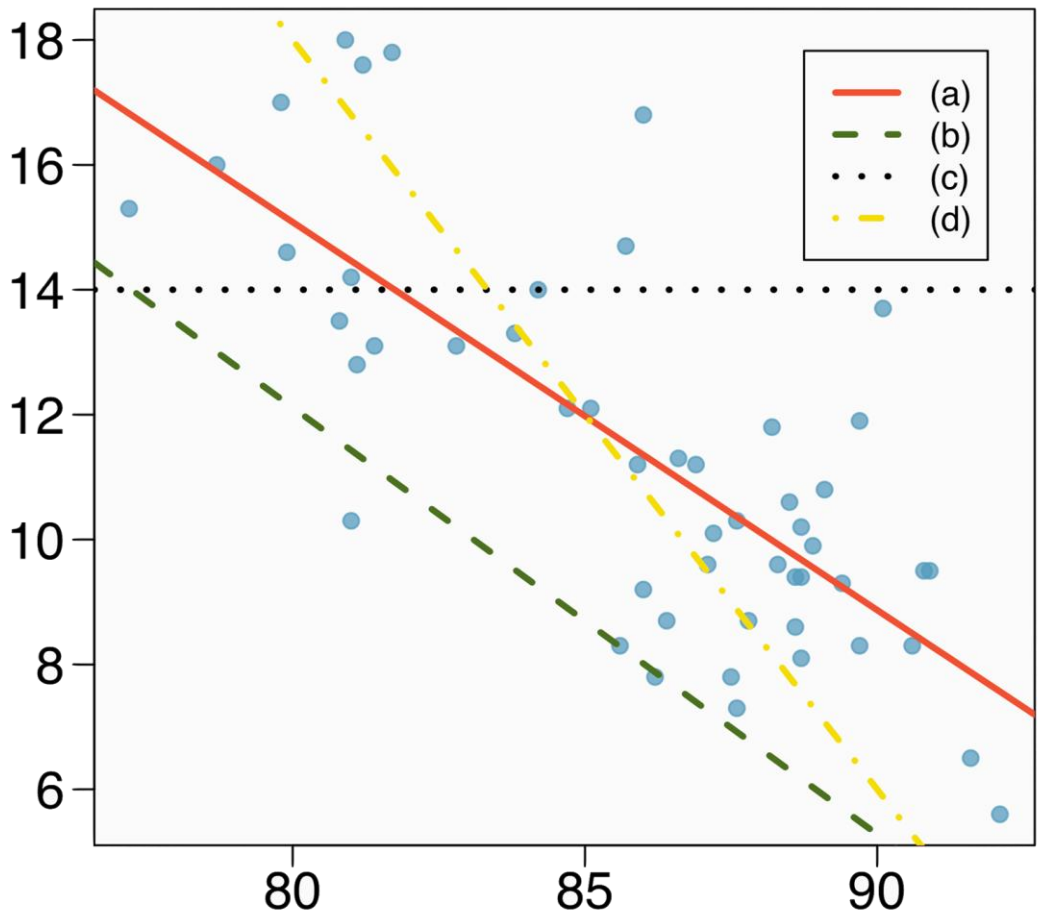
If we fit a line through the data the **vertical differences** between the data and the line are called **residuals**.

Each one of these residuals shows the difference between the real y-coordinate and what the equation of the line predicts from the x coordinate.

Residuals represent the **error** that a prediction would have. **Linear Regression** is a method that involves finding the best $\boldsymbol{\beta_1}$ and $\boldsymbol{\beta_0}$ (i.e. the best equation y = mx + c) to minimise the error.

EYEBALLING THE LINE OF BEST FIT

Which of the following appears to be the line that best fits the data?

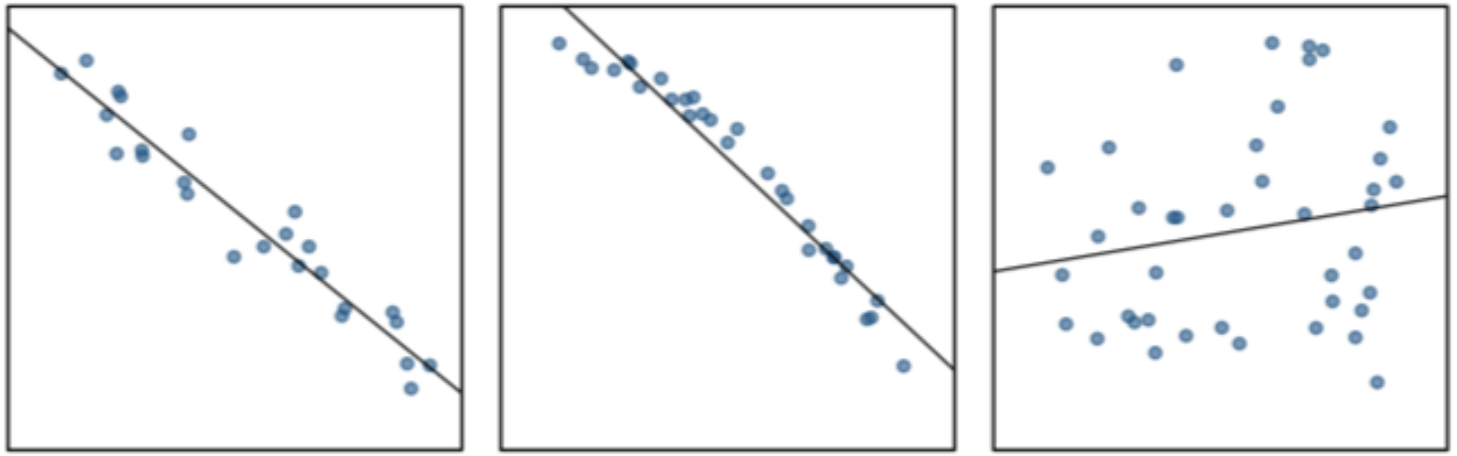


LINEAR REGRESSION CONDITIONS

Anytime that we use technology to generate a line of best fit it will be using linear regression. This is then called a **regression line**.

Most tools will calculate the line regardless of whether it's likely to be a good fit or not – they find the best fit that can be done with a line and rely on human checking.

There are complex conditions for checking – such as linearity, normally distributed residuals, no extreme outliers, and constant variability, but the majority of the time we can simplify this to asking: **is the data elliptical?**

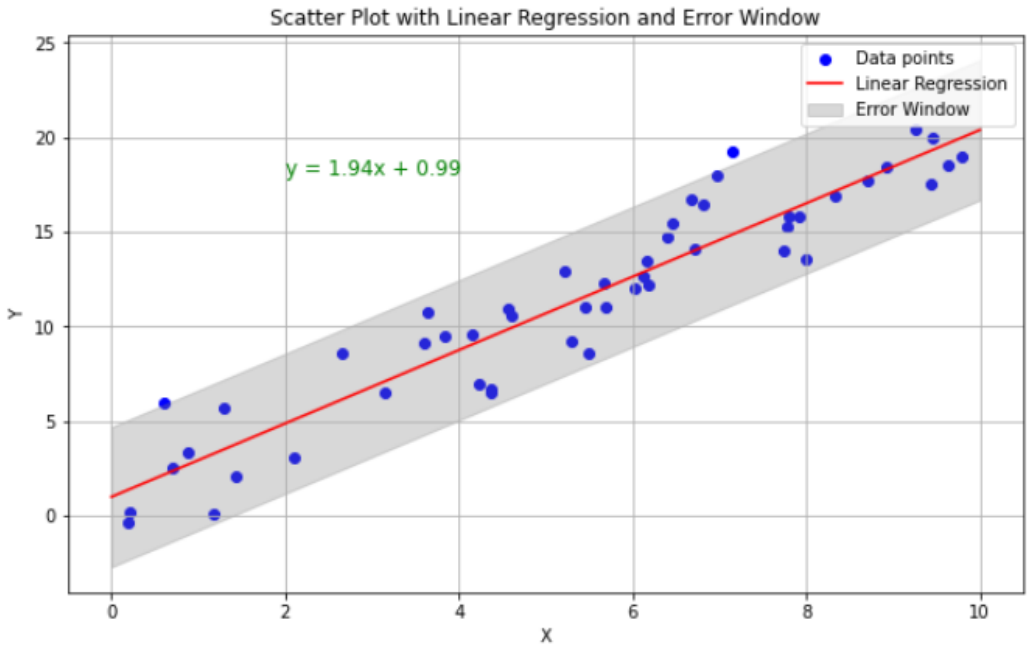


PREDICTING FROM A LINEAR MODEL

The equation for the regression line is calculated using existing data.

Suppose a new value of x is requested and we want to predict the value of y that goes with it.

If x=6, the equation gives us y=12.63, but we can see from the error window that it could be as big as about 17 or as small as about 8.5.

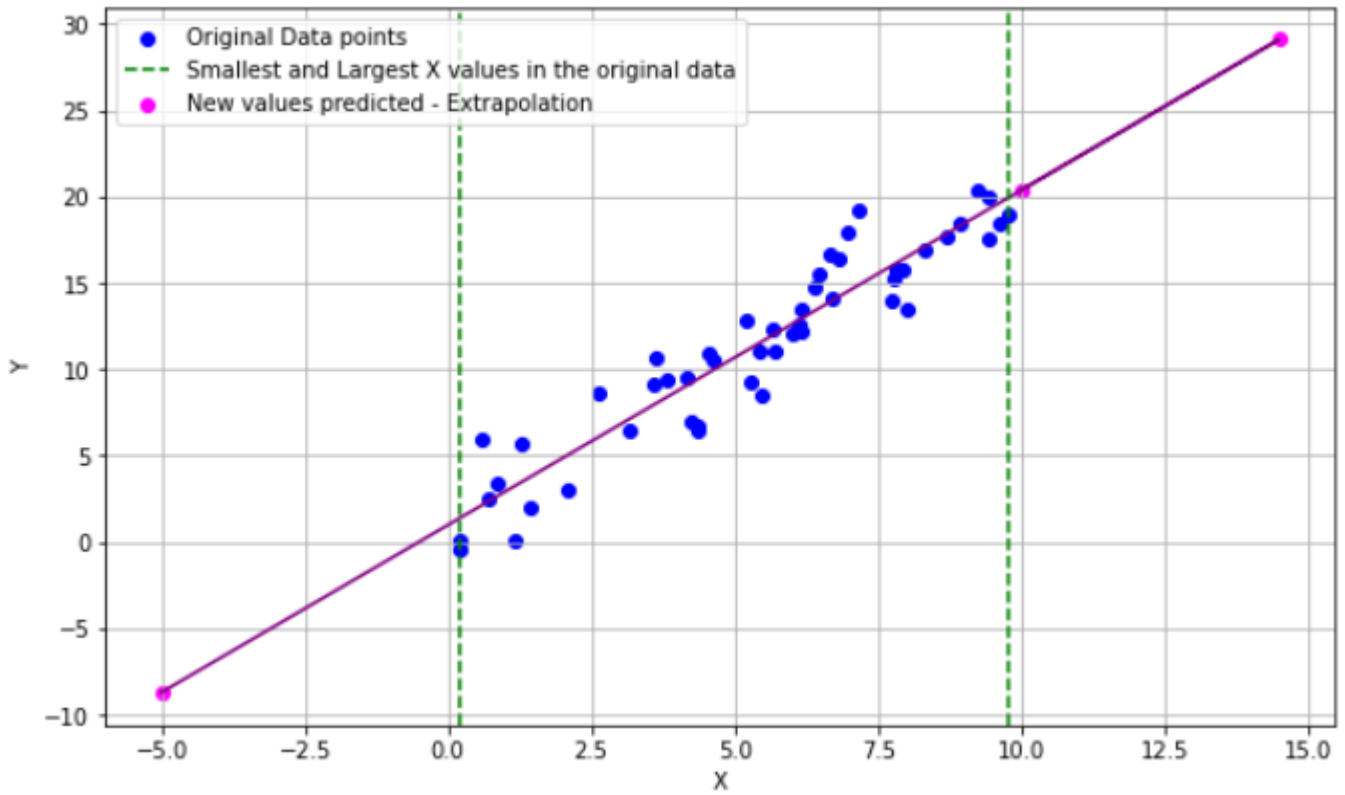


EXTRAPOLATION

Interpolation / Extrapolation

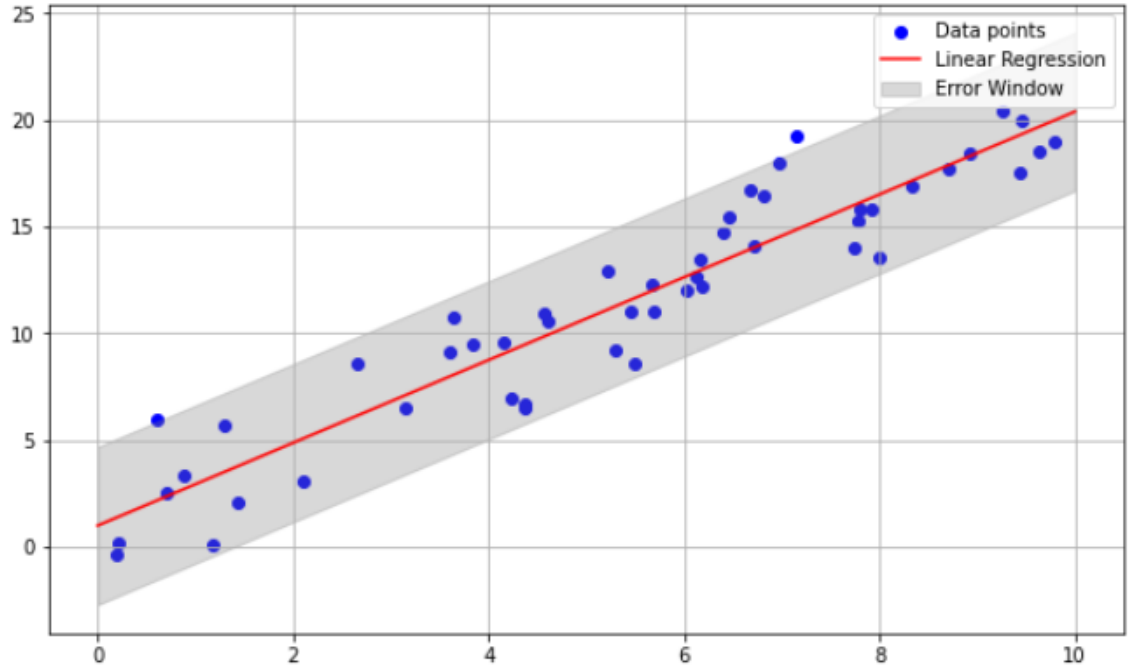
In between the points = reliable

Outside the points = unreliable



NEXT STEPS

In this module we have not followed Machine Learning development practices to fully test and evaluate the model – if you want to find out more, linear regression is often used as the first example on Machine learning courses to illustrate the processes involved.




```
r = 0.94 and p-value = 1.0471603915160772e-24
y = 1.94x + 0.99
R Squared = 0.89
```